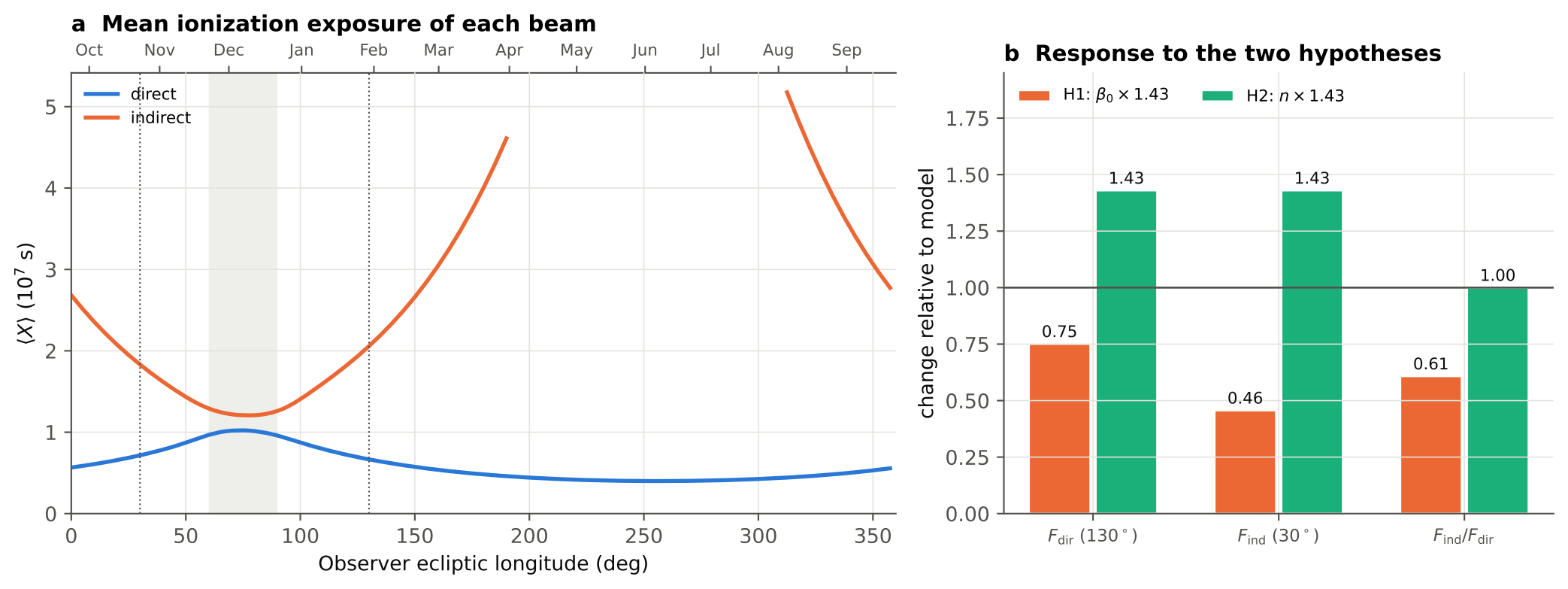
a Mean ionization exposure of each beam
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
direct
indirect
⟨X⟩ (10⁷ s)
0
1
2
3
4
5
0
50
100
150
200
250
300
350
Observer ecliptic longitude (deg)
b Response to the two hypotheses
H1: β₀ × 1.43
H2: n × 1.43
change relative to model
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0.75
1.43
0.46
1.43
0.61
1.00
F_dir (130°)
F_ind (30°)
F_ind/F_dir